

HILP 与候选筛选

1 问题定义与论文符号

依据 Khonji & Khalifa (AAAI 2023) 的定义、引理 3.3 与算法 1-3, 整理 CC-POMDP 版本的核心逻辑; 算法编号沿用原文, 按正文数学定义统一初始化与停止条件, 省略代码实现细节。

$$M = \langle S, A, \mathcal{O}, T, O, U, b_0, h \rangle, \quad M'' = M \parallel \langle \mathcal{R}, \Delta \rangle,$$

$$\max_{\pi} \mathbb{E} \left[\sum_{q \in \tilde{\mathcal{O}}^{\pi}: \tau(q) > \varsigma} U(S_q, \pi(q)) \middle| \pi \right], \quad er(0 | \pi) \leq \Delta.$$

符号	含义
$S, A, \mathcal{O}$	状态、动作、观测的有限集合。
$T(s, a, s'), O(o, s', a)$	状态转移概率、到达 s' 后产生观测 o 的概率。
$U(s, a), b_0, h$	单步期望效用、初始信念、规划时域 (不是动作步数)。
$\mathcal{R}, \Delta$	危险状态集合、执行过程中至少一次进入危险状态的概率上限。
$\pi, \tilde{\mathcal{O}}^{\pi}, er(0 \pi)$	确定性条件策略、其观测历史树、从初始状态起的执行风险 (包含初态)。
$q, 0, qa, qo$	动作-观测历史、空历史、追加动作、追加观测; $q \leq q'$ 表示 q 是 q' 的前缀, 含自身。
$\tilde{A}, \tilde{\mathcal{O}}$	以动作、观测结尾的历史集合; 空历史 $0 \in \tilde{\mathcal{O}}$ 。
$a_q^i, o_q^i, q $	历史中的第 i 个动作、观测及历史步数。
$D(s, a), \tau(q), \varsigma$	动作持续时间、继续执行的判定值、停止阈值; $\tau(q) > \varsigma$ 时继续, 否则停止。
$r(b), R$	$r(b) = \sum_{s \in \mathcal{R}} b(s)$; ILP 剩余风险预算 $R = \Delta - r(b_0)$, 不同于集合 $\mathcal{R}$ 。
u_q, r_q, x_q	动作历史的概率加权效用、首次失败概率贡献、二元选择变量; $x_q = 1$ 表示策略选择该动作历史。
$\mathcal{E}$	已展开的动作历史节点集合, 后续分支已进入搜索范围, 使用精确系数 u_q, r_q 。
$\mathcal{F}$	已生成、尚未继续向下搜索的候选动作历史节点集合; 即 frontier, 不是按效用筛出的集合。
$\mathcal{N}$	待生成下一层候选动作的观测历史集合; 初始为 $\{0\}$ 。
h_q^u, h_q^r	前沿后续效用的可采纳上界、风险的可采纳下界, 均按到达该历史的概率加权。
$\mathbf{x}$	ILP 的二元解, 表示一棵条件策略树, 不是一条动作序列。

当前部分树对应的 p-ILP (全文记为 P) :

$$\max_{x_q \in \{0,1\}} \sum_{q \in \mathcal{E}} u_q x_q + \sum_{q \in \mathcal{F}} h_q^u x_q,$$

$$\text{约束: } \sum_{a \in A} x_a = 1,$$

$$\sum_{a \in A} x_{qoa} = x_q \quad (q \in \mathcal{E}, o \text{ 可达且 } \tau(qo) > \varsigma),$$

$$\sum_{q \in \mathcal{E}} r_q x_q + \sum_{q \in \mathcal{F}} h_q^r x_q \leq R.$$

$\mathcal{F} = \emptyset$ 时得到完整树的 ILP。选中父动作后, 每个可达且需继续的观测分支都必须选择一个后继动作; 停止叶仍计入最后一步奖励和风险。

2 Expand: 更新信念、效用、风险与持续时间

Expand 回答“执行这个动作后会发生什么”，不负责挑选最优动作。根信念为 b_0 ，空历史的发生概率为 1；效用与持续时间用普通信念，风险只累计首次进入危险状态的概率。

Algorithm 2 Expand(qa): CC-POMDP

Input: History q , action a , current beliefs, model M'' , and duration D .

Output: Utility u_{qa} , risk r_{qa} , and reachable observation histories qao with updated beliefs and $\tau(qao)$.

- 1 Predict the possible next states after action a using T
 - 2 Compute u_{qa} : expected one-step utility, weighted by the probability of reaching q
 - 3 Compute r_{qa} : probability of staying safe through q and first entering $\mathcal{R}$ after a
 - 4 **for each** reachable observation o **do**
 - 5 Create observation history qao
 - 6 Update its ordinary and safe-prefix beliefs and history probabilities using O
 - 7 Compute $\tau(qao)$ from the chosen duration model
 - 8 Mark qao as continuing if $\tau(qao) > \varsigma$; otherwise mark it as stopped
 - 9 **end for**
 - 10 **return** u_{qa}, r_{qa} and the observation results
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零安全概率不删除普通观测分支，只令相应风险贡献为零；无需对零概率信念归一化。停止分支仍计入本次动作的奖励与风险，但不再生成后继动作。

持续时间判定（从 Expand 中独立列出）：

符号	含义
$\tilde{b}_q^{i-1}(s), s^{i-1}$	给定完整历史 q ，第 i 个动作开始前状态为 s 的平滑后验，以及该时刻的随机状态；平滑使用普通信念及后续观测。
$\mu_{s,a}, \sigma_{s,a}^2$	$D(s, a)$ 的均值与方差；S 模式假设独立高斯噪声。

$$\mu_q = \sum_{i=1}^{|q|} \sum_s \tilde{b}_q^{i-1}(s) \mu_{s,a_q^i}, \quad \sigma_q^2 = \sum_{i=1}^{|q|} \sum_s [\tilde{b}_q^{i-1}(s)]^2 \sigma_{s,a_q^i}^2.$$

符号	含义
Fixed	$D(s, a) = c_a$: $\tau(q) = h - \sum_{i=1}^{ q } c_{a_q^i}$, $\varsigma = 0$ 。
Expected (E)	$\tau(q) = h - \mu_q$, $\varsigma = 0$ 。
Stochastic (S)	$\tau(q) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{h - \mu_q}{\sqrt{2}\sigma_q} \right) \right]$; $\sigma_q = 0$ 时按 $\mu_q < h$ 取 1，否则取 0。
状态依赖确定性时长	若使用分位数准则，则 $\tau(q) = \Pr(\sum_i D(s^{i-1}, a_q^i) < h \mid q)$ ，通过状态与累计时长的增广分布计算，而非仅取均值。

3 Preprocess 与 HILP：完整展开和选择性展开

Algorithm 1 Preprocess(M'', D, ς)

Input: Durative CC-POMDP and stopping threshold.

Output: Full-tree coefficients (u_q, r_q) , budget R , and duration function τ .

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1 Initialize the root from  $b_0$ ;  $R \leftarrow \Delta - r(b_0)$ ;  $\mathcal{N} \leftarrow \{0\}$ ;  $\mathcal{E} \leftarrow \emptyset$ 
2 while  $\mathcal{N} \neq \emptyset$  do
3   Take one pending observation history  $q$  out of  $\mathcal{N}$ 
4   for each  $a \in A$  do
5     Call Expand( $qa$ ); store its results and add  $qa$  to  $\mathcal{E}$ 
6     Add every continuing observation history  $qao$  to  $\mathcal{N}$ 
7   end for
8 end while
9 return  $(u_q, r_q)_{q \in \mathcal{E}}, R, \tau$ 

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Algorithm 3 HILP($M'', D, \varsigma, \text{Solver}$)

Input: Durative CC-POMDP, admissible heuristics h^u, h^r , and an ILP solver.

Output: A policy $\mathbf{x}$, or infeasibility.

```

1 Initialize the root from  $b_0$ ;  $R \leftarrow \Delta - r(b_0)$ 
2  $\mathcal{E} \leftarrow \emptyset$ ;  $\mathcal{F} \leftarrow \emptyset$ ;  $\mathcal{N} \leftarrow \{0\}$ 
3 repeat
4   Generate frontier: create  $qa$  for every  $q \in \mathcal{N}$  and  $a \in A$ 
5   Call Expand( $qa$ ) for each new candidate; store its results and add it to  $\mathcal{F}$ 
6   Assign bounds  $h_{qa}^u, h_{qa}^r$ ; use exact  $u_{qa}, r_{qa}$  if all reachable observations stop
7    $\mathcal{N} \leftarrow \emptyset$ 
8   Select a partial policy:  $\mathbf{x} \leftarrow \text{Solver}[\text{p-ILP}(\mathcal{E}, \mathcal{F})]$ 
9   if p-ILP is infeasible then return infeasibility
10  Extend the selected frontier:
11  for each  $q \in \mathcal{F}$  with  $x_q = 1$  do
12    Move  $q$  from  $\mathcal{F}$  to  $\mathcal{E}$ ; replace its bounds by exact  $u_q, r_q$ 
13    Add every reachable  $qo$  with  $\tau(qo) > \varsigma$  to  $\mathcal{N}$ 
14  end for
15  Keep all unselected action-history nodes in  $\mathcal{F}$ 
16 until  $\mathcal{N} = \emptyset$ 
17 return  $\mathbf{x}$ 

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符号	含义
Preprocess	不看效用排名，展开所有需继续的历史；随后 Full-ILP 一次求解完整模型。
Expand 与 frontier	Expand 计算一步结果；候选动作历史节点进入 $\mathcal{F}$ ，被 ILP 选中后才继续生成后继节点。
中间解 $\mathbf{x}$	$x_q = 1$ 决定下轮向哪里展开；处理全部被选中的前沿，不是只选效用最大的一个。
停止与最优性	假设持续时间停止条件使搜索树有限。停止时 $\mathcal{N} = \emptyset$ ，不要求 $\mathcal{F} = \emptyset$ ；最优性依赖可采纳界及精确求解各 p-ILP。

4 前缀固定 / Prefix Fixing

中文符号表

符号	含义
$\mathcal{E}, \mathcal{F}$	当前树中已展开、前沿的动作历史节点集合；元素 $q \in \tilde{A}$ 是以动作结尾的动作-观测历史。当前部分模型为 p-ILP($\mathcal{E}, \mathcal{F}$)。
$\mathbf{x}^-, \mathbf{x}, \mathbf{x}^+$	上一轮、本轮、下一轮的 Gurobi 解； $\mathbf{x}^{(i)}$ 为第 i 轮的解，含重建的固定变量。新节点在旧解中的分量记为 0；解不一定已是完整策略。
$\mathcal{H}, n$	保存最近 $n \geq 2$ 轮 ILP 解向量的队列；满时 $\mathcal{H} = (\mathbf{x}^{(t-n+1)}, \dots, \mathbf{x}^{(t)})$ 。由外层 HILP 维护，初始为空，不足 n 轮时保留已有结果，满后先进先出，相同结果不去重。
$C_1, C_0; C_1^+, C_0^+$	本轮及下一轮固定选择/排除的动作历史节点集合，初始为空；固定的是对应节点的 $x_q = 1/0$ ，不是基本动作 a 。
$q \leq q'$	q 是动作历史 q' 的前缀，包含 $q = q'$ ；用于排除整个分支。
$\perp$	Gurobi 无可返回解时的返回值，不代表原问题无解；仍返回 C_1^+, C_0^+ 。有解时先取得自由变量，再补回固定分量。

English symbol table

Symbol	Meaning
$\mathcal{E}, \mathcal{F}$	Expanded and frontier sets of action-history nodes. Each $q \in \tilde{A}$ is an action-observation history ending in an action. The current partial model is p-ILP($\mathcal{E}, \mathcal{F}$).
$\mathbf{x}^-, \mathbf{x}, \mathbf{x}^+$	Previous, current, and next Gurobi solutions; $\mathbf{x}^{(i)}$ is the solution at iteration i , including reconstructed fixed variables. New nodes have zero entries in older solutions. A solution need not be a complete policy.
$\mathcal{H}, n$	A queue of the last $n \geq 2$ ILP solution vectors; when full, $\mathcal{H} = (\mathbf{x}^{(t-n+1)}, \dots, \mathbf{x}^{(t)})$. Maintained by outer HILP, initially empty; retain available entries until full, then evict the oldest. Keep duplicate results.
$C_1, C_0; C_1^+, C_0^+$	Current and next sets of action-history nodes fixed as selected/excluded, initially empty. The fixings $x_q = 1/0$ apply to individual history nodes, not primitive actions a .
$q \leq q'$	Action history q is a prefix of q' , including equality; used to exclude a whole branch.
$\perp$	Gurobi returns $\perp$ if no solution is available, not proof of original infeasibility; still return C_1^+, C_0^+ . Otherwise, recover fixed entries after solving for free variables.

Algorithm 4 PrefixFixing: $t \rightarrow t+1$ (replaces line 11 of the original HILP algorithm)

Input: $(\mathcal{E}, \mathcal{F}, \mathcal{H}, n, C_1, C_0)$.

Output: $(\mathbf{x}^+, C_1^+, C_0^+)$.

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1   $C_1^+ \leftarrow C_1; C_0^+ \leftarrow C_0$ 
2  if  $|\mathcal{H}| = n$  then
3     $C_1^+ \leftarrow \mathcal{E} \cup \mathcal{F}$ 
4    for each  $\mathbf{x}^{(i)} \in \mathcal{H}$  do
5       $C_1^+ \leftarrow \{q \in C_1^+ : x_q^{(i)} = 1\}$ 
6    end for
7     $C_1^+ \leftarrow C_1 \cup C_1^+$ 
8  end if
9   $C_0^+ \leftarrow C_0^+ \cup \{qa' : qa \in C_1^+, a' \in A \setminus \{a\}\}$ 
10  $C_0^+ \leftarrow \{q' \in \mathcal{E} \cup \mathcal{F} : \exists q \in C_0^+, q \leq q'\}$ 
11 if  $C_1^+ \cap C_0^+ \neq \emptyset$  then return  $(\perp, C_1^+, C_0^+)$ 
12  $\mathbf{x}^+ \leftarrow \text{Gurobi}(\text{p-ILP}(\mathcal{E}, \mathcal{F})|_{x_q=1 (q \in C_1^+), x_q=0 (q \in C_0^+)})$ 
13 if  $\mathbf{x}^+ = \perp$  then return  $(\perp, C_1^+, C_0^+)$ 
14 for each  $q \in C_1^+$  do
15    $x_q^+ \leftarrow 1$ 
16 end for
17 for each  $q \in C_0^+$  do
18    $x_q^+ \leftarrow 0$ 
19 end for
20 return  $(\mathbf{x}^+, C_1^+, C_0^+)$ 

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// Gurobi binaries after elimination: $|\mathcal{E} \cup \mathcal{F}| \rightarrow |\mathcal{E} \cup \mathcal{F}| - |C_1^+ \cup C_0^+|$; global optimality is not guaranteed.

5 心跳筛选 / Heartbeat

中文符号表

符号	含义
$\mathcal{E}, \mathcal{F}$	当前树中已展开、前沿的动作历史节点集合；元素 $q \in \tilde{A}$ 是以动作结尾的动作-观测历史。当前部分模型为 $\text{p-ILP}(\mathcal{E}, \mathcal{F})$ 。
$\mathbf{x}^-, \mathbf{x}, \mathbf{x}^+$	上一轮、本轮、下一轮的解；用 $\mathbf{x}$ 判断最近选中的动作历史节点。首次调用及新节点取 $x_q = 0$ ，仅表示此前未选中。
ω, ω^+	本轮及下一轮按 q 索引的心跳字典；外层 HILP 在首次调用前置 $\omega \leftarrow \emptyset$ ，后续传入上轮返回值，不再清空。 $q \notin \text{dom } \omega$ 表示新节点。
ω_0, λ	重置分数 $\omega_0 \geq 1$ 、衰减因子 $0 < \lambda < 1$ ；下一轮分数小于 1 的未保护分支被排除。
C_0, C_0^+	本轮及下一轮累计排除的动作历史节点集合，初始为空；代入 $x_q = 0$ ($q \in C_0^+$) 后消元，保留根约束及全部观测流约束。
$q \leq q'$	前缀关系，含自身；未排除的新节点、本轮选中节点及其祖先不被剪除，但不强制其 $x_q = 1$ 。
$\perp$	Gurobi 无可返回解时返回 $\perp$ ；不证明原问题无解。

English symbol table

Symbol	Meaning
$\mathcal{E}, \mathcal{F}$	Expanded and frontier sets of action-history nodes. Each $q \in \tilde{A}$ is an action-observation history ending in an action. The current partial model is $\text{p-ILP}(\mathcal{E}, \mathcal{F})$.
$\mathbf{x}^-, \mathbf{x}, \mathbf{x}^+$	Previous, current, and next solutions; $\mathbf{x}$ identifies selected action-history nodes. Initially, and for new nodes, $x_q = 0$ only marks no previous selection.
ω, ω^+	Current and next heartbeat maps indexed by q . Outer HILP initializes $\omega \leftarrow \emptyset$ before the first call, then passes the previous output without clearing it. Nodes outside $\text{dom } \omega$ are new.
ω_0, λ	Reset score $\omega_0 \geq 1$ and decay factor $0 < \lambda < 1$; exclude unprotected branches whose next score is below 1.
C_0, C_0^+	Current and next sets of excluded action-history nodes, initially empty. Eliminate $x_q = 0$ ($q \in C_0^+$) while retaining root and all observation-flow constraints.
$q \leq q'$	Prefix relation, including equality. Protect unexcluded new/currently selected nodes and their ancestors from pruning, without fixing $x_q = 1$.
$\perp$	Gurobi returns $\perp$ if no solution is available; this does not prove original infeasibility.

Algorithm 5 Heartbeat: $t \rightarrow t + 1$ (replaces line 11 of the original HILP algorithm)

Input: $(\mathcal{E}, \mathcal{F}, \mathbf{x}, \omega, C_0, \omega_0, \lambda)$.

Output: $(\mathbf{x}^+, \omega^+, C_0^+)$.

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1   $C_0^+ \leftarrow \{q' \in \mathcal{E} \cup \mathcal{F} : \exists q \in C_0, q \leq q'\}; \omega^+ \leftarrow \omega$ 
2  for each  $q \in \mathcal{E} \cup \mathcal{F}$  do
3    if  $q \notin \text{dom } \omega$  then  $\omega_q^+ \leftarrow \omega_0$ 
4    if  $q \notin C_0^+ \wedge (q \notin \text{dom } \omega \vee x_q = 1)$  then  $\omega_q^+ \leftarrow \omega_0$ 
5    else  $\omega_q^+ \leftarrow \lambda \omega_q^+$ 
6    end if
7  end for
8   $C_0^+ \leftarrow C_0^+ \cup \{q \in \mathcal{E} \cup \mathcal{F} : \omega_q^+ < 1,$ 
       $\nexists q' \in (\mathcal{E} \cup \mathcal{F}) \setminus C_0^+ :$ 
       $q \leq q' \wedge (q' \notin \text{dom } \omega \vee x_{q'} = 1)\}$ 
9   $C_0^+ \leftarrow \{q' \in \mathcal{E} \cup \mathcal{F} : \exists q \in C_0^+, q \leq q'\}$ 
10  $\mathbf{x}^+ \leftarrow \text{Gurobi}(\text{p-ILP}(\mathcal{E}, \mathcal{F})|_{x_q=0 (q \in C_0^+)})$ 
11 if  $\mathbf{x}^+ = \perp$  then return  $(\perp, \omega^+, C_0^+)$ 
12 for each  $q \in C_0^+$  do
13    $x_q^+ \leftarrow 0$ 
14 end for
15 return  $(\mathbf{x}^+, \omega^+, C_0^+)$ 

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// Gurobi binaries after eliminating C_0^+ : $|\mathcal{E} \cup \mathcal{F}| \longrightarrow |\mathcal{E} \cup \mathcal{F}| - |C_0^+|$; global optimality is not guaranteed.

6 Rank-based: 自底向上的子树筛选（保留比例）

先对全部 $\mathcal{F}$ 打分并保留前 $\lceil \lambda |\mathcal{F}| \rceil$ 个作为 $\mathcal{F}'$ ；再从 $(\mathcal{F} \setminus \mathcal{F}')$ 的父动作节点自底向上收集候选子树根 $\mathcal{E}_{\text{subtree-root}}$ 。若集合中所有候选子树未保护的 E 数量之和仍不足剩余删除目标 $\lceil \lambda |\mathcal{E}| \rceil$ 对应的缺口，则不评分、不试删，直接把 $\mathcal{E}_{\text{subtree-root}}$ 上移到它们的父节点继续。一旦容量达标，才按子树 Score 从低到高删除可删部分；删除时只移除未被保护、且删除后同一观测组仍有备选的完整动作侧枝，**始终保留包含 $\mathcal{F}'$ 的那条路径**。若删完本层仍未达标，则继续上移一层。与当前 Python 实现一致；比例是软目标，不保证最优性。

中文符号与操作定义（算法 6）

符号或操作	含义
$\mathcal{E}, \mathcal{F}; \mathcal{E}', \mathcal{F}'$	输入的已展开、未展开动作历史集合；输出的保留子集。中间的 $\mathcal{F}'$ 是初选高分集合，返回前更新为实际保留的全部 F。
$q, q', o; G, n$	动作历史、后继动作历史、观测；任意动作历史子集 $G \subseteq \mathcal{E} \cup \mathcal{F}$ 、非负整数 n 。
$u_q, r_q; h_q^u, h_q^r$	输入的 E 精确系数、F 启发式系数，已包含历史到达概率权重。
α, λ, R	输入参数：风险评分权重 $\alpha \geq 0$ 、保留比例 $0 < \lambda \leq 1$ 、剩余风险预算 $R = \Delta - r(b_0)$ 。
$\mathbf{x}^-, \text{DeadEnds}$	输入的上一轮 ILP 二元解、已知无合法后续策略的动作历史集合；首次调用或未出现的变量取 $x_q^- = 0$ 。
$\text{kept}, \text{protected}$	当前保留的动作历史集合；禁止删除的历史集合，包括初选 F、仍可用的上一轮选择及其全部祖先。
$\mathcal{E}_{\text{subtree-root}}, \text{parents}$	候选子树根集合；该集合的父动作历史中仍在 $\mathcal{E}'$ 的部分，供下一轮上移。
$\text{subtreeSize}(q)$	q 子树中仍保留、未保护的 E 数量；包括已完成的 E 叶节点，不包含 F。
$\text{target}, \text{capacity}, \text{cut}$	本轮剩余 E 删除目标；候选子树并集中未保护 E 的数量（可删量上界）；一次实际删除的动作历史集合。
$\cap, \cup, \setminus, \emptyset;$ $ G , \lceil \cdot \rceil$	交集、并集、差集、空集；集合大小、向上取整。例如 $\mathcal{F} \cap \text{kept}$ 是当前仍保留的 F，不是乘法。
$\text{Parents}(G),$ $\text{Ancestors}(G)$	G 的最近父动作历史集合（跳过观测历史）；G 的全部严格祖先动作历史集合。根动作没有父动作；空集合返回空集合。
$\text{Subtree}(q)$	输入树中以 q 为根的所有后代动作历史，包含 q 自身。
$\text{Observations}(q),$ $\text{Actions}(qo)$	q 后可达且需继续执行的观测；观测历史 qo 后的备选动作历史集合，即一个 OR 动作组。根动作组为 $\text{Actions}(0)$ 。
$\text{Usable}(G, R)$	返回 G 中仍可达、局部可用的历史：排除 DeadEnds 和前缀风险系数之和超过 $R + 10^{-6}$ 的节点；自底向上排除任一必要观测组没有可用动作的节点，再自顶向下排除其后代。前缀风险使用 E 的 r_q 与 F 的 h_q^r ，包含自身；不证明全局可行性。
$\text{Top}(G, n),$ $\text{Sort}(G, \text{Score} \uparrow)$	按 Score 降序取至多 n 个历史，返回集合；按 Score 升序返回 G 的有序列表（ $\uparrow$ 表示升序）。同分使用输入树的固定动作遍历顺序。
$\text{OuterRoots}(G)$	返回 $\{q \in G : \text{Ancestors}(\{q\}) \cap G = \emptyset\}$ ，删除嵌套候选根，使候选子树互不重叠。
$\text{Removable}(q, \text{kept}, \text{protected}, n)$	返回可删侧枝并集：只检查含未保护 E 的存活子树，按固定动作顺序自顶向下遍历。仅当节点未保护且同一 OR 组还有其他存活动作时整棵删枝，否则检查其孩子。每次试删更新局部视图；删去的 E 数量达到 n 即停止（整棵删枝可超额）。不删 protected ；无可删 E 或 q 已删除时返回空集。
$\text{TrimFrontier}(\text{kept}, \text{protected})$	返回更新后的 kept：逐个存活 OR 组删除未保护 F；若该组全部是未保护 F，则留下最高分的一个。同分按固定动作顺序。

English symbols and operation definitions (Algorithm 6)

Symbol or operation	Definition
$\mathcal{E}, \mathcal{F}; \mathcal{E}', \mathcal{F}'$	Input expanded/frontier action histories; retained output subsets. Intermediate $\mathcal{F}'$ is the initial high-score selection; final $\mathcal{F}'$ contains all retained frontier histories.
$q, q', o; G, n$	An action history, a successor action history, an observation; any history subset $G \subseteq \mathcal{E} \cup \mathcal{F}$ and a nonnegative integer n .
$u_q, r_q; h_q^u, h_q^r$	Input exact E coefficients and heuristic F coefficients, already weighted by history probabilities.
α, λ, R	Input risk-score weight $\alpha \geq 0$, retention fraction $0 < \lambda \leq 1$, and residual risk budget $R = \Delta - r(b_0)$.
$\mathbf{x}^-, \text{DeadEnds}$	Input previous binary ILP solution and known histories with no legal continuation. Set $x_q^- = 0$ on the first call or for an unseen variable.
kept, protected	Currently retained histories; histories forbidden to be deleted: initially selected $\mathcal{F}$, usable previous choices, and all their ancestors.
$\mathcal{E}_{\text{subtree-root}}, \text{parents}$	Candidate subtree roots; their parent action histories still in $\mathcal{E}'$, used in the next upward round.
subtreeSize(q)	Retained, unprotected E count in the subtree of q , including completed E leaves but excluding F.
target, capacity, cut	Remaining E deletion quota; unprotected E count in the union of candidate subtrees (an upper bound); histories actually removed in one cut.
$\cap, \cup, \setminus, \emptyset;$ $ G , \lceil \cdot \rceil$	Intersection, union, set difference, empty set; cardinality and ceiling. $\mathcal{F} \cap \text{kept}$ means frontier histories still retained.
Parents(G), Ancestors(G)	Nearest parent action histories (skip observation histories); all strict ancestor action histories of G . Root actions have no parent action; an empty input gives an empty set.
Subtree(q)	All descendant action histories in the input tree, including q .
Observations(q), Actions(qo)	Reachable observations requiring continuation after q ; alternative action histories after qo (one OR group). Actions (0) is the root action group.
Usable(G, R)	Return locally usable, reachable histories: exclude DeadEnds and histories whose prefix risk exceeds $R + 10^{-6}$; bottom-up, exclude a history if any required observation group has no usable action; top-down, exclude its descendants. Prefix risk sums E's r_q and F's h_q^r , including the current history. This is not a global feasibility certificate.
Top(G, n), Sort($G, \text{Score} \uparrow$)	Return a set of up to n highest-score histories; return G as an ascending-score list. Resolve ties by the fixed input-tree action traversal order.
OuterRoots(G)	Return $\{q \in G : \text{Ancestors}(\{q\}) \cap G = \emptyset\}$, removing nested roots to obtain disjoint subtrees.
Removable(q , kept, protected, n)	Return deletable side branches. Inspect only surviving subtrees containing unprotected E, top-down in fixed action order. Delete a whole unprotected action subtree only if its OR group has another surviving action; otherwise inspect its children. Update the local view after each deletion; stop after deleting n E histories (a whole branch may overshoot). Never delete protected histories. Return empty if q is absent or no E can be deleted.
TrimFrontier(kept, protected)	Return updated kept after removing unprotected F in each surviving OR group; keep its highest-score F if all available actions are unprotected F. Use fixed action order for ties.

$$\text{Score}(q) = \begin{cases} h_q^u - \alpha h_q^r, & q \in \mathcal{F}, \\ u_q - \alpha r_q + \sum_{o \in \text{Observations}(q)} \max_{q' \in \text{Actions}(qo) \cap \text{kept}} \text{Score}(q'), & q \in \mathcal{E}. \end{cases} \quad (1)$$

Score is evaluated bottom-up on kept, children before parents. An empty observation sum is zero; Usable ensures every required maximum has at least one action. Candidate scores are computed once per scoring round, before that round's cuts. / Score 在 kept 上按孩子先于父亲的顺序计算; 空和为零, Usable 保证每个必要观测的最大值集合非空。同一评分轮次先统一评分, 再按该评分删枝。

Algorithm 6 BottomUpRankPruning($\mathcal{E}, \mathcal{F}, \mathbf{x}^-; \alpha, \lambda, R$)

Input: $\mathcal{E}, \mathcal{F}$, their coefficients and tree links (with fixed action order), DeadEnds, previous solution $\mathbf{x}^-$, and parameters α, λ, R .

Output: $\mathcal{E}' \subseteq \mathcal{E}, \mathcal{F}' \subseteq \mathcal{F}$.

```

1  if  $\lambda = 1$  then return  $(\mathcal{E}, \mathcal{F})$ 
2  for each  $q \in \mathcal{F}$  do Score( $q$ )  $\leftarrow h_q^u - \alpha h_q^r$ 
3  kept  $\leftarrow$  Usable( $\mathcal{E} \cup \mathcal{F}, R$ );  $\mathcal{E}' \leftarrow \mathcal{E} \cap$  kept
4   $\mathcal{F}' \leftarrow$  Top( $\mathcal{F} \cap$  kept,  $\lceil \lambda |\mathcal{F}| \rceil$ )
5  protected  $\leftarrow \mathcal{F}' \cup \{q \in \text{kept} : x_q^- = 1\}$ 
6  protected  $\leftarrow$  protected  $\cup$  Ancestors(protected)
7   $\mathcal{E}_{\text{subtree-root}} \leftarrow$  Parents( $(\mathcal{F} \cap \text{kept}) \setminus \text{protected}$ )  $\cap \mathcal{E}'$ 
8   $\mathcal{E}_{\text{subtree-root}} \leftarrow \mathcal{E}_{\text{subtree-root}} \cup \{q \in \mathcal{E}' : \text{Observations}(q) = \emptyset\}$ 
9  while  $|\mathcal{E}'| > \lceil \lambda |\mathcal{E}| \rceil$  and  $\mathcal{E}_{\text{subtree-root}} \neq \emptyset$  do
10    $\mathcal{E}_{\text{subtree-root}} \leftarrow$  OuterRoots( $\mathcal{E}_{\text{subtree-root}} \cap$  kept)
11   target  $\leftarrow |\mathcal{E}'| - \lceil \lambda |\mathcal{E}| \rceil$ 
12   for each  $q \in \mathcal{E}_{\text{subtree-root}}$  do
13     subtreeSize( $q$ )  $\leftarrow |(\text{Subtree}(q) \cap \mathcal{E}') \setminus \text{protected}|$ 
14   end for
15   capacity  $\leftarrow \sum_{q \in \mathcal{E}_{\text{subtree-root}}} \text{subtreeSize}(q)$ 
16   parents  $\leftarrow$  Parents( $\mathcal{E}_{\text{subtree-root}}$ )  $\cap \mathcal{E}'$ 
17   if capacity  $> 0$  and (capacity  $\geq$  target or parents  $= \emptyset$ ) then
18     Evaluate Score( $q$ ) for all  $q \in$  kept bottom-up using (1)
19     for each  $q \in \text{Sort}(\mathcal{E}_{\text{subtree-root}}, \text{Score} \uparrow)$  do
20       cut  $\leftarrow$  Removable( $q$ , kept, protected,  $|\mathcal{E}'| - \lceil \lambda |\mathcal{E}| \rceil$ )
21       kept  $\leftarrow$  kept  $\setminus$  cut;  $\mathcal{E}' \leftarrow \mathcal{E} \cap$  kept
22       if  $|\mathcal{E}'| \leq \lceil \lambda |\mathcal{E}| \rceil$  then break
23     end for
24   end if
25    $\mathcal{E}_{\text{subtree-root}} \leftarrow$  parents  $\cap \mathcal{E}'$ 
26 end while
27 kept  $\leftarrow$  TrimFrontier(kept, protected)
28  $(\mathcal{E}', \mathcal{F}') \leftarrow (\mathcal{E} \cap \text{kept}, \mathcal{F} \cap \text{kept})$ 
29 return  $(\mathcal{E}', \mathcal{F}')$ 

```

// Removable returns empty for a removed root. Recompute each cut after earlier deletions.

// Count before scoring; capacity is only an upper bound because observation coverage may require keeping extra histories.

// Accumulate subtreeSize bottom-up; after deletion, update affected ancestors. Do not enumerate every subtree repeatedly.

// At the highest reachable layer, apply the remaining valid cuts even if the quota is unattainable.

// Submitted binaries: $|\mathcal{E} \cup \mathcal{F}| \rightarrow |\mathcal{E}' \cup \mathcal{F}'|$; the ratio is soft and optimality is not guaranteed.

原树不修改; 仅用 $\mathcal{E}', \mathcal{F}'$ 构建受限 ILP, 未保留变量取零。Gurobi 求解与受限不可行时的回退仍在算法外。/ The input tree is unchanged; build the restricted ILP on $\mathcal{E}', \mathcal{F}'$ with omitted variables fixed to zero. Solving and infeasibility fallback remain outside this procedure.